FLCR-04 - D4, J3(O), Triality, And Representation Transport

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper registers the eight-state LCR carrier across multiple lossless local presentations: LCR triples, an axis/sheet codec, and a diagonal J3(O)-style carrier. The result is a finite coordinate atlas, not a claim that full exceptional dynamics have been solved. Later papers may import representation transport only by preserving the registered readouts or declaring the dropped readout.

Keywords

D4; J3(O); finite atlas; axis sheet; representation transport.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **D4, J3(O), Triality, And Representation Transport**. The paper's immediate contribution is: Places representation changes into finite chart and J3(O) transport discipline.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. D4, J3(O), Triality, And Representation Transport.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Defines the finite atlas connecting LCR, axis/sheet, and diagonal carrier readings.
- Separates diagonal J3(O) registration from full off-diagonal exceptional algebra.
- Introduces representation-invariance as a corpus import rule.
- Routes full D4/F4/off-diagonal claims to later evidence packages.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- `03_D4_J3_Triality_Surface.md`
- `supplements/03_INTERNAL_REASONING_SUPPLEMENT.md`

Cross-corpus evidence classes:

- `CAM_CLAIM_100_SCORE_AUDIT.md`
- `NSIT_TOOL_CLOSURE_MATRIX.md`
- `NSIT_REASONED_CLOSURE_PASS.md`
- `PAPER_REWORKS_CRYSTAL_PROJECTION.json`
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Finite coordinate atlas.** A set of lossless readouts over the same finite state set with verified transitions.
- **Axis/sheet codec.** An antipodal quotient plus a sheet lift that recovers the selected state.
- **Diagonal J3(O) carrier.** A diagonal-only matrix-style readout avoiding off-diagonal octonion multiplication burdens.
- **Representation-invariance rule.** A claim survives transport only when its required readouts agree or the omitted readout is explicitly bounded.

4. Formal Claims

Claim	Lane	Statement
Theorem 4.1	`receipt_bound_internal_result`	The eight binary LCR states admit a lossless finite atlas across LCR, axis/sheet, and diagonal carrier coordinates.
Proposition 4.2	`normal_form_result`	The axis/sheet codec is an antipodal quotient followed by a sheet lift.
Protocol 4.3	`falsifier_or_open_obligation`	Full D4, F4, or off-diagonal J3(O) assertions are downstream claims unless a separate action receipt is attached.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 4.1	finite/executable result	The eight binary LCR states admit a lossless finite atlas across LCR, axis/sheet, and diagonal carrier coordinates	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 4.2	formal construction	The axis/sheet codec is an antipodal quotient followed by a sheet lift	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 4.3	obligation/falsifier	Full D4, F4, or off-diagonal J3(O) assertions are downstream claims unless a separate action receipt is attached	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	`FLCR-04` defines or uses **D4, J3(O), Triality, And Representation Transport** as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Fix the eight-state carrier from FLCR-02 as the shared state set.

2. Define each allowed readout as a coordinate chart over that same set.
3. Check round-trip preservation through LCR, axis/sheet, and diagonal carrier readings.
4. Forbid downstream imports that use full exceptional algebra language without the extra action receipt.

Proof Sketch

The proof is an atlas proof over a finite set. A readout is admitted only when it maps each of the eight states to a unique code and the inverse map recovers the original state. Because the diagonal carrier uses only diagonal idempotent-style data, it does not require the unproved off-diagonal octonion multiplication burden.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is `D4, J3(O), Triality, And Representation Transport`. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Paper 03 legacy body	D4/J3/triality surface, chart registration, D12 and block-tower imports.
Internal supplement 03	Atlas language, diagonal-only guard, representation-invariance rule.
NSIT tool matrix	T1-T4 algebra foundation, T5-T7 closure, D12 idempotent chain, D4 block tower receipts.
CAM Claim 100 Score Audit	Paper 03 scored 94 internally closed, with full D4/F4/off-diagonal theorem routed to NP work.

Appendix C. Implementation Intake

This addendum binds implementation work that was not fully represented in the earlier paper routing. The rows below are not pasted source text; they are evidence-lane imports that change what the paper can claim now.

Implementation source	Lane	Claim effect	Boundary
IMPL-R30-LATTICE-THEOREM-REGISTRY	`standard_theorem_citation_bound_result`	Adds executable theorem registry rows for octonions, J3(O), chart-to-J3(O), exact n=3 SU(3) Weyl closure, Rule-30 8x8 transition, Niemeier routes, relational qubit closure, Monster/Pariah boundary, and computational ISA signatures. Verification: Targeted pytest passed: 11 Rule-30 normal-form/proof-shell tests, including address decomposition, centroid state, proof-shell dependency acyclicity, Niemeier obligation boundaries, scoped claim promotion, and full dependency closure.	Imports verifier-backed internal/formal results; transported physical interpretations remain lane-bound.

The immediate paper effect is stronger claim posture where these implementation rows satisfy the relevant lane. Remaining caution is reserved for specific claims that still lack an adapter, comparator, calibration target, or rerun receipt.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	`receipt_bound_internal_result`	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.511321.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
representation fold	`normal_form_result`	The fold/transport action is closed as a representation map with named invariants and bounded residues.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-1 tower	`receipt_bound_internal_result`	The decade tower is resolved with avg TarPit mass=0.510370 and mass sum=40.829567.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-04 cross-lift comparison	`cam_crystal_reapplication_result`	Family tower binds FLCR-04, FLCR-14, FLCR-24, FLCR-34; avg TarPit mass=0.508927; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
D4/J3(O)/triality carrier	`standard_theorem_citation_bound_result`	Representation-transport vocabulary is closed where imported algebra is cited and invariants are preserved.	Physical particle identity requires translation/comparator rows.

Claim Posture After This Pass

`FLCR-04` should now be read as a resolved-state contribution for `D4, J3(O), Triality, And Representation Transport` at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- `FLCR-11` through `FLCR-18` for window, receipt, admission, CA, algebraic,

curvature, residue, and exceptional machinery.

- `FLCR-28` through `FLCR-30` for CAM/crystal, corpus ribbon, and universal

normal-form intake.

- `FLCR-31` through `FLCR-40` only after the LCR-native result has been cited

and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- Diagonal registration does not prove full exceptional dynamics.
- A codec migration must preserve round-trip bijection over the eight states.
- Later Standard Model translation papers cannot consume this as physical transport without calibration.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): `03`/fold_action.

Original slot	Family	Lift	Role	Proof form
`03`	fold_action	0	order-1 slot-3: fold the initial action into a higher-dimensional representation	fold map, coordinate atlas, and representation transport

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 02 and reopened at original slot 03 (D4 J3 Triality Surface).

It must produce: formal result of original slot 03 plus the same-family lift path toward slot 13.

Semantic transition: fold the local state into a higher representation while preserving addressability.

Accepted formalism targets to bind in the journal rewrite:

- representation theory
- group actions and equivariance
- tensor or coordinate atlases
- transport maps between equivalent descriptions

Slot	Family	Lift	Produced proof form
`03`	fold_action	0	fold map, coordinate atlas, and representation transport

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to strengthen representation transport: every face, gauge, atlas, or fold label must point to a preserved source object.

Primary Evidence Inputs

Source	Placement reason
`03_D4_J3_Triality_Surface.md`	primary assigned source for the paper's formal spine
`supplements/03_INTERNAL_REASONING_SUPPLEMENT.md`	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations

Secondary And Orbital Evidence Inputs

Source	Placement reason
`supplements/RECEIPT_VERIFIER_CATALOG.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/LATTICE_FORGE_UNIFICATION_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_PYRAMID_INDEX.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`CAM_CLAIM_100_SCORE_AUDIT.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`NSIT_TOOL_CLOSURE_MATRIX.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`NSIT_REASONED_CLOSURE_PASS.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`ORBITAL_DATA_CROSS_POPULATION_AUDIT.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`AGENT_CRYSTAL_WORK_HARVEST.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`PAPER_REWORKS_CRYSTAL_PROJECTION.json`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

03	94	internally closed	LCR/D4/J3 registration, T1-T7, D12, S3 anneal, D4 atlas, block tower	full D4/F4/off-diagonal theorem is NP-04

Paper-Specific NSIT Closure Rows

T1-T4 algebra foundation	batch_tool_unison_closure	`octonion.py`, `jordan_j3.py`, `rule30.py`, `f4_action.py`	Paper 03	`algebra_foundation_T1_T4_receipt.json` 8/8	Octonions, J3(O), chart isomorphism, exact n=3 closure bound.
T5-T7 plus shell-2 doublet	batch_tool_unison_closure	`f4_action.py`, `core.py`	Papers 03 and 01	`su3_closure_T5_T7_receipt.json` 10/10; `bijective_shell2_doublet_receipt.json` 7/7	SU(3)/8x8 closure and single-tape doublet close.
T_D12 idempotent chain	individual_tool_closure	`lattice_forge/d12_action.py`	Paper 03	`d12_idempotent_chain_receipt.json` 6/6	D12 idempotent chain closes on D4 chart.
D4 block tower + G2->F4 exceptional conjugate	batch_tool_unison_closure	`block_d4.py`, `block_tower.py`, `g2_f4_t5_conjugate.py`	Paper 03	`d4_block_tower_exceptional_receipt.json` 3/3	D4 block, block tower, and exceptional conjugate close.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Promote now	Bind next	Residue
none directly assigned	Source routing and claim lanes identify paper-local binding actions.	Verifier catalogs and CAM/crystal routes provide object-name evidence candidates.	Unbound claims remain in falsifier/open-obligation lanes.

Binding Discipline

Each queue item is represented as a delta:

```
local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
```

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Search By Object Name	No direct reasoned-closure queue was assigned from the first queue map.	Search source routing, verifier catalog, CAM/crystal routes, and paper-local object names.	Create a specific binding queue when an object-name match finds a validator, receipt, theorem, or calibration route.	Until then, claims remain in their existing lanes.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-04.25: Paper 4.25 - Toolkit for Boundary Repair	81	Paper 4.25 describes the review tools for boundary repair. The tools help a reader construct and inspect repair rows. They do not add claims beyond the Paper 4 theorem. The toolkit works with: ```text correction state...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FINAL_FORMAL_PAPER: Complete Formal Claims: The Folded Strand	75	We present the complete closed-form claim set of the CQE_CMPLX corpus: - 33 papers = 33 folding operations on a self-complementary strand - 144 tools = cumulative analog kit = digital twin surface - 135 digital twins ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.

Evidence card	Score	Publication contribution	Boundary
SUMMARY-V-32-Theorems-Registry: Summary Paper V — The 32 Theorems in One Table: Closed-Form Registry	75	This paper is the **complete closed-form registry of all 32 theorems** in the CQE_CMPLX corpus. Each theorem is stated precisely, given its formal context (where it is proven), and listed with its verifier. The table ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-02.50: Paper 2.50 - Correction Surface Claim Contract	73	Paper 2.50 lets later papers import the correction surface honestly. It keeps the finite theorem available while preserving the distinction between residue, obligation, and proof.	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
A7_HONESTY: Appendix A7: Honesty Policy & Compositional Closure	71	/ Code / Meaning / /------/---------/ / **PASS** / All checks pass / / **PASS_WITH_OPEN_LIFTS** / Some checks open but no failures / / **FAIL** / Any check fails / / **PARTIAL** / Some pass, some fail / / **BOUNDED_EX...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	26 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	`FLCR-04`
Legacy source slot(s)	`03`
Ribbon role	order-1 slot-3: fold the initial action into a higher-dimensional representation
Proof form	fold map, coordinate atlas, and representation transport
Received state	state emitted by prior slot 02 and reopened at original slot 03 (D4 J3 Triality Surface)
Produced state	formal result of original slot 03 plus the same-family lift path toward slot 13

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 4.1	`receipt_bound_internal_result`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	The eight binary LCR states admit a lossless finite atlas across LCR, axis/sheet, and diagonal carrier coordinates.
Proposition 4.2	`normal_form_result`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	The axis/sheet codec is an antipodal quotient followed by a sheet lift.
Protocol 4.3	`falsifier_or_open_obligation`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Full D4, F4, or off-diagonal J3(O) assertions are downstream claims unless a separate action receipt is attached.

Figure Plan

Figure	Purpose	Caption
FLCR-04-F1	Representation fold atlas	Schematic showing how `FLCR-04` turns its received state into the produced state while preserving claim lanes and residue boundaries.

Figure	Purpose	Caption
FLCR-04-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-04-F3	Same-family lift context	Diagram placing this paper in the original `00-79` ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-04-F1: Representation fold atlas

```
received state
  -> Representation fold atlas
  -> formal claim lane
  -> evidence object / receipt / citation
  -> produced state
  -> remaining residue
```

Table Plan

Table	Purpose
FLCR-04-T1	Face/representation correspondence table
FLCR-04-T2	Claim-lane/evidence-boundary table
FLCR-04-T3	Archive evidence card and source-backed upgrade table
FLCR-04-T4	Workbook replay and falsifier table

Worked Example

Map one source object across two labels/faces while preserving its address. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
atlas	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
face	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
fold	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.
representation transport	Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-04-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

`FLCR-04` (`D4, J3(O), Triality, And Representation Transport`) occupies the `fold_action` role at lift depth `0`. The paper receives state emitted by prior slot 02 and reopened at original slot 03 (D4 J3 Triality Surface). It produces formal result of original slot 03 plus the same-family lift path toward slot 13. The state transition is: fold the local state into a higher representation while preserving addressability.

The paper-local proof form is:

```
fold map, coordinate atlas, and representation transport
```

The integration result is a representation-transport chart with named invariants and residues. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
representation theory	Formal carrier for the paper-local result.
group actions and equivariance	Formal carrier for the paper-local result.
tensor or coordinate atlases	Formal carrier for the paper-local result.
transport maps between equivalent descriptions	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 4.1	`receipt_bound_internal_result`	The eight binary LCR states admit a lossless finite atlas across LCR, axis/sheet, and diagonal carrier coordinates.
Proposition 4.2	`normal_form_result`	The axis/sheet codec is an antipodal quotient followed by a sheet lift.
Protocol 4.3	`falsifier_or_open_obligation`	Full D4, F4, or off-diagonal J3(O) assertions are downstream claims unless a separate action receipt is attached.

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	`03_D4_J3_Triality_Surface.md`, `supplements/03_INTERNAL_REASONING_SUPPLEMENT.md`	routed evidence
Evidence inputs	`CAM_CLAIM_100_SCORE_AUDIT.md`, `NSIT_TOOL_CLOSURE_MATRIX.md`, `NSIT_REASONED_CLOSURE_PASS.md`, `PAPER_REWORKS_CRYSTAL_PROJECTION.json`	routed evidence
CAM/crystal sources	`PAPER_REWORKS_CRYSTAL_PROJECTION.json`, `AGENT_CRYSTAL_WORK_HARVEST.json`, `runtime standards artifact`, `notebook-derived source bundle`	routed evidence
Standards-window inputs	`window_summary.json`, `node DBs`, `window DBs`	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting

the operation;

4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-04-F1: Representation fold atlas	Visualizes the proof object, routing, or boundary.
FLCR-04-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-04-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-04-T1: Face/representation correspondence table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-04-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-04-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-04-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.